Boundary conditions

Frederik Dalby

Modelling Fall 2025

Three speecial cases with Neumann where flux (mass or heat) = 0

1) Heat or mass cannot get out (closed or insulated in the end)
2) The end is open and nothing is described about what happens (no convection description)
3) There is symmetry typically at index 0 (could be e.g. in middle of a sphere or cylinder)

Three overall types

Boundary Condition	Mathematical Form	Physical Meaning	
Dirichlet	$C = \text{given value}$	Fixed concentration at boundary	(first type)
Neumann	$\frac{\partial C}{\partial x} = \text{given value}$	Fixed flux (including zero flux)	(second type)
Robin	$aC + b\frac{\partial C}{\partial x} = \text{given value}$	Mixed condition; flux proportional to concentration difference	(third type)

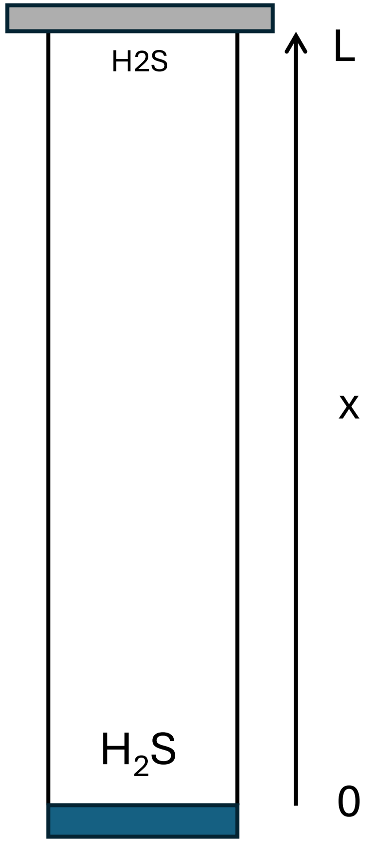
```
# Dirichlet boundary conditions for a mass transfer problem
# at last index (index N)
# if initial condition at index N is also the boundary condition then:
dCdt[N] = 0

# if boundary is given at the ghost point then, the value (C_right in this case)
# is passed to the rates function first.
dCdt[N] = D * (C[N-1] - 2*C[N] + C_right)/dx**2
```

Use constitutive equations at boundary (newtons law, mass transfer coefficient, reaction kinetics)

We can have different type of boundary conditions in each end: Example with Dirichlet + Neumann BC

A well with contaminated water at the bottom is emitting H_2S. There is always 1000 ppm H_2S at the bottom. On the top of the well a concrete lid ensures that H_2S cannot escape.



Neumann BC (fixed flux)

$$J(t, x = L) = -D\frac{\partial C_{(t,x=L)}}{\partial x} = 0$$

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

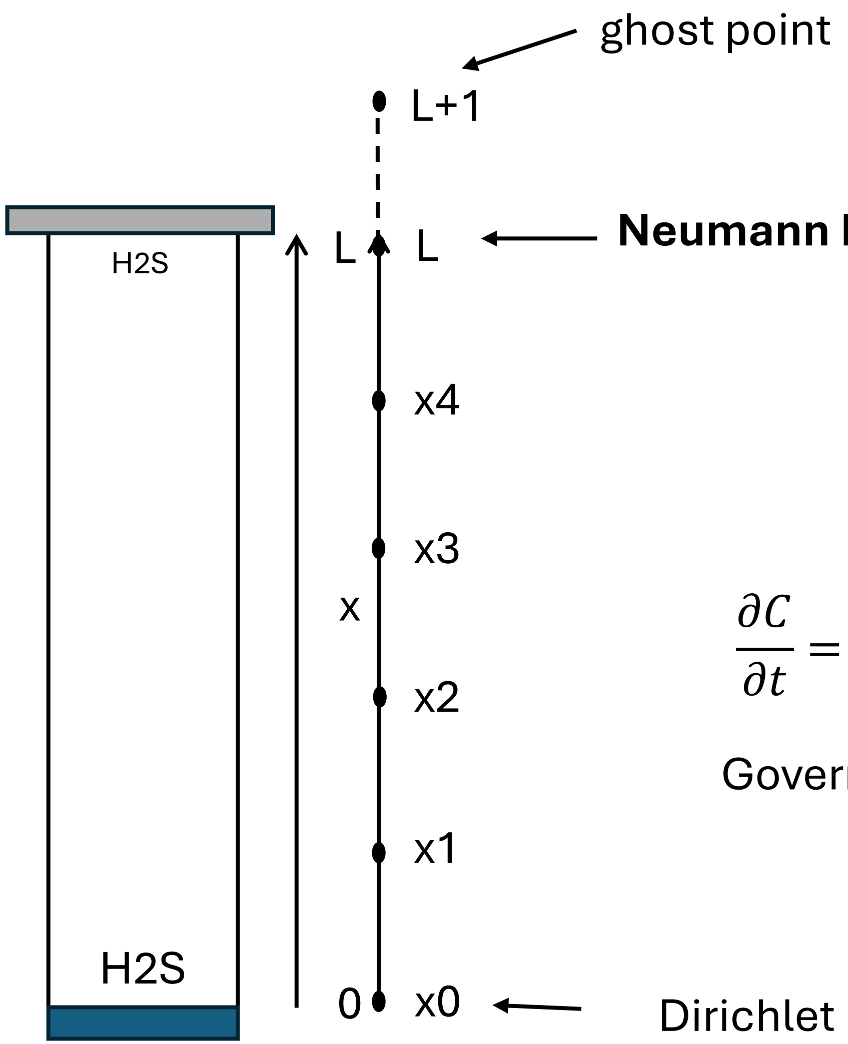
Governing eq

2 BC (for all time points)

1 IC (for every ODE)

Dirichlet BC

$$C(t, x = 0) = 1000$$



Neumann BC (fixed flux)

$$J = -D \frac{\partial C_{(t,z=L)}}{\partial x} = 0$$

$$\frac{\partial C}{\partial x} = 0$$

$$\frac{c_{L+1} - c_{L-1}}{2\Delta x} = 0$$

$$c_{L+1} = c_{L-1}$$

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

Governing eq

Dirichlet BC

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$ Governing eq

reformulate to discretized form

$$\frac{\partial C}{\partial t} = D\frac{c_{i-1} - 2c_i + c_{i+1}}{\Delta x^2}$$

At the top of the well

$$\frac{\partial C(t, x = L)}{\partial t} = D\frac{c_{L-1} - 2c_L + c_{L+1}}{\Delta x^2}$$

Substitute $c_{L+1} = c_{L-1}$

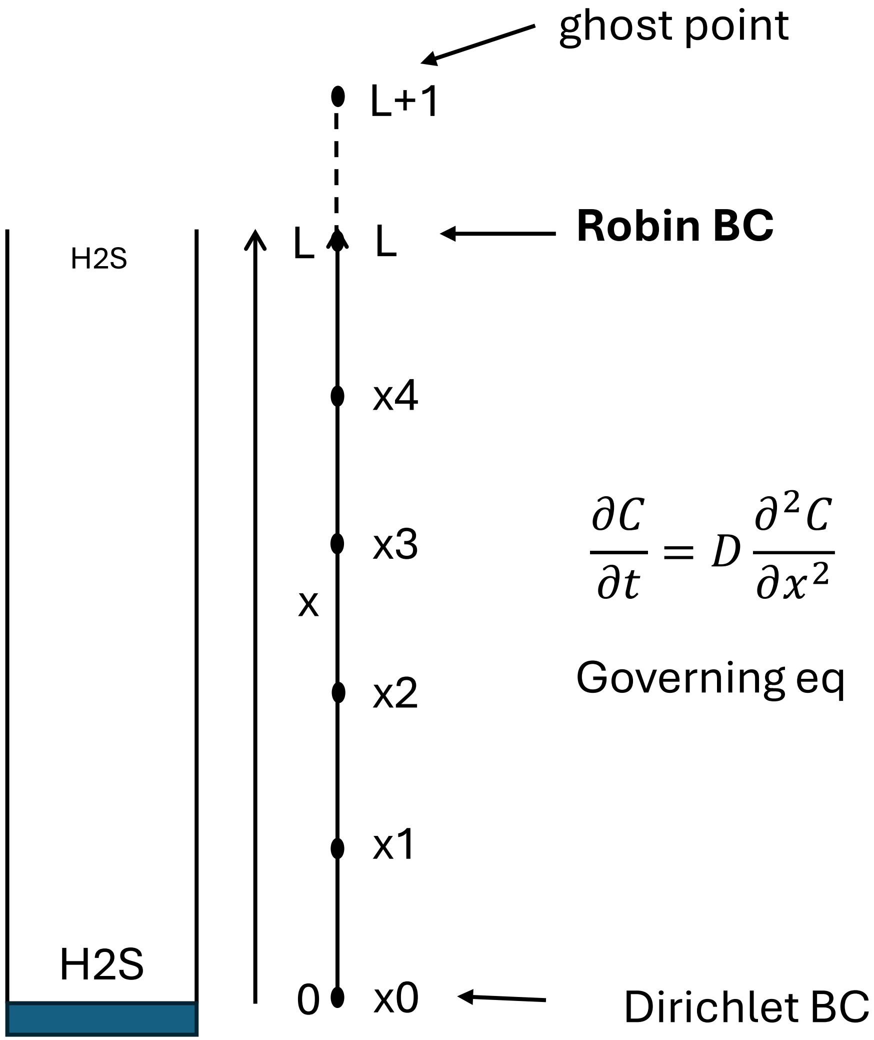
$$\frac{\partial C(t, x = L)}{\partial t} = D\frac{c_{L-1} - 2c_L + c_{L-1}}{\Delta x^2} = D\frac{2(c_{L-1} - c_L)}{\Delta x^2}$$

Code for Neumann BC with dcdx = 0 at last index (N)

```
# J = -D*dc/dx = 0: -D * (c[i+1] - c[i-1])/(2*dx) = 0
# At boundary:
# -D * (c[N+1] - c[N-1])/(2*dx) = 0: c[N+1] is the ghost point
# c[N+1] = c[N-1]

dcdt[N] = D * (c[N-1] - 2 * c[N] + c[N-1])/dx**2
# can also be written like this:
dcdt[-1] = D * (c[-2] - 2 * c[-1] + c[-2])/dx**2
```

A well with contaminated water at the bottom is emitting H_2S. There is always 1000 ppm H_2S at the bottom. The top of the well is open to the air with a mild wind over the well. H_2S is transferred to the bulk air phase by convection



$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

Governing eq

$$J = -D\frac{dC_{(t,x=L)}}{dx} = f(C_{(t,x=L)}, C_{bulk})$$

$$-D\frac{dC(t, x = L)}{dx} = k_c \cdot (C_{(t,x=L)} - C_{bulk})$$

$$-D\frac{c_{L+1} - c_{L-1}}{2\Delta x} = k_c \cdot (C_L - C_{bulk})$$

$$c_{L+1} - c_{L-1} = \frac{2\Delta x k_c \cdot (C_L - C_{bulk})}{-D}$$

$$c_{L+1} = -\frac{2\Delta x k_c \cdot (C_L - C_{bulk})}{D} + c_{L-1}$$

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$ Governing eq

↓ reformulate to discretized form

$$\frac{\partial C}{\partial t} = D\frac{c_{i-1} - 2c_i + c_{i+1}}{\Delta x^2}$$

↓ At the top of the well

$$\frac{\partial C(t, x = L)}{\partial t} = D\frac{c_{L-1} - 2c_L + c_{L+1}}{\Delta x^2}$$

Substitute $$c_{L+1} = -\frac{2\Delta x k_c \cdot (C_L - C_{bulk})}{D} + c_{L-1}$$

$$\frac{\partial C(t, x = L)}{\partial t} = D\frac{c_{L-1} - 2c_L + (-\frac{2\Delta x k_c \cdot (C_L - C_{bulk})}{D} + c_{L-1})}{\Delta x^2} = 2D\frac{c_{L-1} - c_L}{\Delta x^2} - 2k_c\frac{c_L - c_{bulk}}{\Delta x}$$

code for heat transfer at last index (N)

```
# Robin BC the right side because it is open to the air and
# we can assume that the heat is transfered to outside the model domain
# by convection. The heat flux law is Fouriers law and at the boundary it
# is balanced by the heat convection (Newtons law). The boundary is index N
# Below Fouriers law = Newtons law
# q = -k*dT/dx = h * (T[N] - T_air)
# q = -k * (T[N+1]- T[N-1])/(2*dx) = h * (T[N] - T_air)
# isolate T[N+1]:
# T[N+1] = T[N-1] - 2dx*h/k * (T[N] - T_air)
# write ghost point variable here:
TN1 = T[N-1] - 2*dx*h/k * (T[N] - T_air)
dTdt[N] = alpha * (TN1 - 2*T[N] + T[N-1])/dx**2
```

code for mass transfer at last index (N)

```
# Robin BC with mass transfer from top of well to open air
# ficks law controls flux at the boundary but it is balanced by convection
# to the outside air.
# below: Ficks law  = mass transfer approach eqaution
# J = -D * dC/dx = k_m * (C[N] - C_air)
# J = -D * (C[N+1]- C[N-1])/(2*dx) = k_m * (C[N] - C_air)
# isolate C[N+1]:
# C[N+1] = C[N-1] - 2dx*k_m/D * (C[N] - C_air)
# write ghost point variable here:
CN1 = C[N-1] - 2*dx*k_m/D * (C[N] - C_air)
dCdt[N] = D * (CN1 - 2*C[N] + C[N-1])/dx**2
```